



ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា



ដេប៉ាតឺម៉ង់ ទេពកោសល្យ អគ្គិសនី និងថាមពល

គម្រោងសញ្ញាបត្រវិស្វកម្ម

ការបង្កើនគុណភាពនៃ Controller MPC និង MPPI ដោយ

ប្រើប្រាស់ Deep Natural Network ក្នុងបរិស្ថានប្រែប្រួល

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ឯកទេស	: អគ្គិសនី និងថាមពល
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**INSTITUT DE TECHNOLOGIE DU CAMBODGE
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MEMOIRE DE FIN D'ETUDES INGENIEUR**

**Réseau Neuronal Profond Basé sur un Modèle pour Améliorer les
Contrôleurs MPC et MPPI dans des Environnements Dynamiques.**

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ក្រសួងអប់រំ យុវជន និងកីឡា

វិទ្យាស្ថានបច្ចេកវិទ្យាភ្នំពេញ



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គម្រោងសញ្ញាបត្រវិស្វកម្ម

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ថ្ងៃទី ខែ ឆ្នាំ ២០២៤

**ការបង្កើនគុណភាពនៃ Controller MPC និង MPPI ដោយ
ប្រើប្រាស់ Deep Natural Network ក្នុងបរិស្ថានប្រែប្រួល**

សហគ្រាស : វិទ្យាស្ថានបច្ចេកវិទ្យាភ្នំពេញ

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Directeur de l'Institut : _____

Phnom Penh, le 2024

**Réseau Neuronal Profond Basé sur un Modèle pour Améliorer les Contrôleurs
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PHNOM PENH, 2024

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Thank the people who supported your thesis work: family, your institution's leadership, your department head, your advisor, and anyone else who helped. One paragraph per person or group works well, as shown in this example structure.

Thank your family or sponsors for their support.

Thank your institution's director/leadership, e.g. **Dr. FIRSTNAME LASTNAME**, Director of *Your Institution*.

Thank your department head and faculty, e.g. **Dr. FIRSTNAME LASTNAME**, Head of *Your Department*.

Thank your thesis advisor, e.g. **Mr./Ms./Dr. FIRSTNAME LASTNAME**.

Thank friends, peers, or anyone else who helped along the way.

សេចក្តីសង្ខេប

សូមសរសេរសេចក្តីសង្ខេបនៃការស្រាវជ្រាវរបស់អ្នកជាភាសាខ្មែរនៅទីនេះ។ អក្សរខ្មែរដែលសរសេរនៅក្នុងបរិយាកាស khmer នេះនឹងបង្ហាញបានត្រឹមត្រូវ។

RESUME

Write your thesis summary in French here. This section is optional; remove it if your institution does not require a French summary.

ABSTRACT

Write your thesis abstract in English here: the research problem, your approach, and your key results and conclusions. One or two short paragraphs is typical.

ABBREVIATIONS AND SYMBOLS

Abbreviations:

ABC	Example abbreviation
XYZ	Another example abbreviation

Symbols:

x	Example variable
α	Example parameter

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1 Introduction

Start your introduction here. Explain the background of your problem, why it matters, and what gap in existing work your thesis addresses.

1.1 Background

Describe the context and prior work related to your topic. Cite sources using `\cite{}`, for example [1].

1.2 Problem Statement

State clearly what problem your thesis solves.

1.3 Objectives

List the objectives of your research, e.g. using an itemized list:

- First objective.
- Second objective.
- Third objective.

1.4 Thesis Structure

Briefly outline how the rest of the thesis is organized (one sentence per chapter/section is typical).

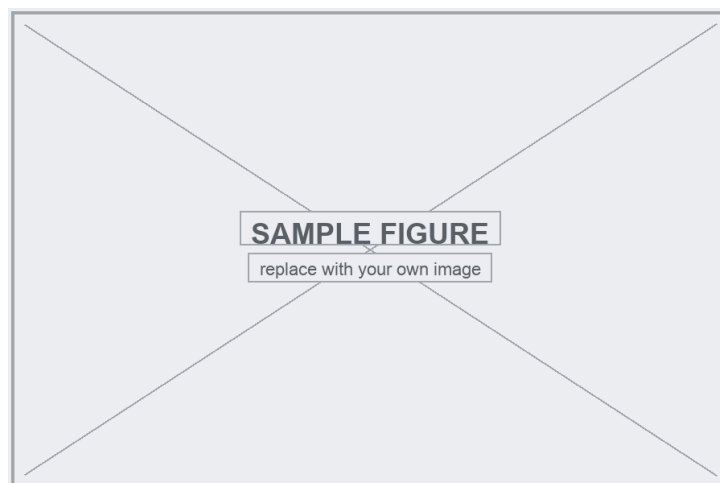


Figure 1.1 Example figure – replace with a diagram of your own, e.g. an overview of your thesis concept

2 Methodology

2.1 Overview

Describe your overall approach or system design here.

2.2 Example: Equation

Mathematical formulations use standard `amsmath` / `unicode-math` syntax, for example a state-space model:

$$\dot{x}(t) = f(x(t), u(t)) \quad (1)$$

where x is the state vector and u is the control input vector.

2.3 Example: Figure

Figures are numbered per section (e.g. Figure 1.1 in Section 1) and listed in the List of Figures automatically.

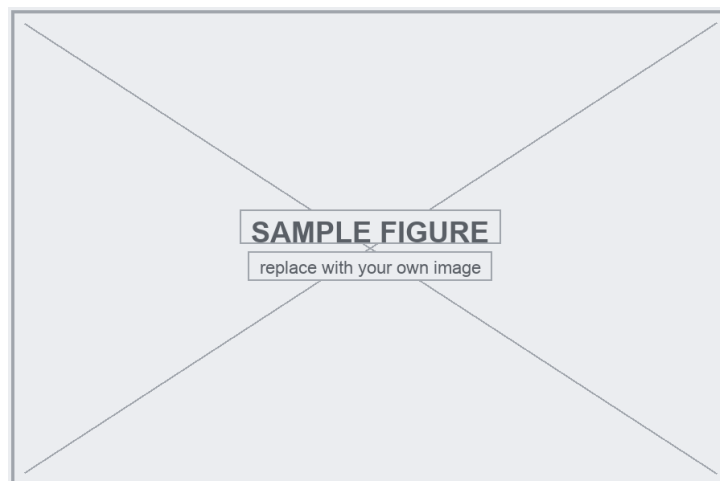


Figure 2.1 Example figure – replace with your own diagram or plot

2.4 Example: Table

Tables follow the same numbering convention as figures.

Table 2.1 Example parameter table

Parameter	Value
Example parameter A	1.0
Example parameter B	0.5

2.5 Example: Algorithm

Pseudocode uses the `algorithm/algpseudocode` packages.

Algorithm 1 Example Algorithm

```
1: Initialize variables
2: for  $i \leftarrow 1$  to  $N$  do
3:   Perform an update step
4: end for
5: return result
```

2.6 Example: Code Block

Python code blocks use the custom `pythoncode` environment (syntax highlighting via `minted`).

```
def example_function(x, u):
    """Replace with your own code example."""
    return x + u
```

3 Results and Discussion

3.1 Experimental Setup

Describe your experimental setup, dataset, or simulation environment here.

3.2 Results

Present your results here, referring to figures and tables as needed, e.g. see Figure 3.1 and Table 3.1.

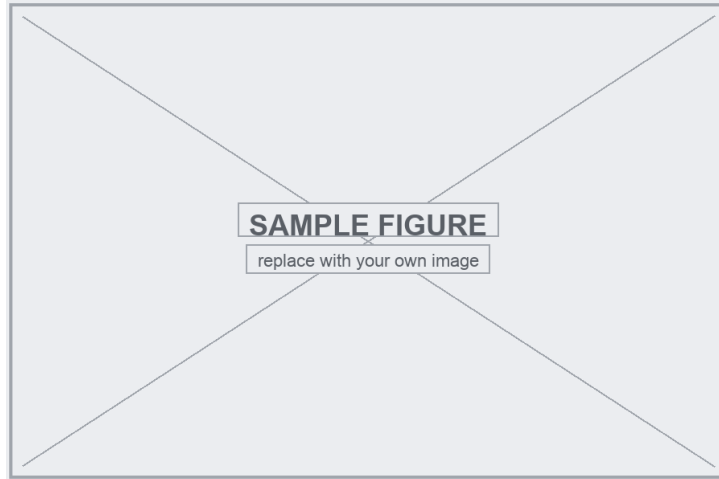


Figure 3.1 Example results figure – replace with your own plot

Table 3.1 Example results comparison

Method	Metric A	Metric B
Baseline	0.00	0.00
Proposed	0.00	0.00

3.3 Discussion

Interpret your results here: what do they mean, how do they compare to prior work (cite as needed, e.g. [2]), and what are the limitations?

4 Conclusion and Future Work

Summarize your thesis here: what you did, what you found, and how it addresses the objectives stated in the introduction.

Discuss the limitations of your work and suggest directions for future research.

References

- [1] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, *Information theoretic model predictive control: Theory and applications to autonomous driving*, 2017. arXiv: 1707.02342 [cs.R0].
- [2] T. Salzmann, E. Kaufmann, J. Arrizabalaga, M. Pavone, D. Scaramuzza, and M. Ryll, “Real-time neural mpc: Deep learning model predictive control for quadrotors and agile robotic platforms,” *IEEE Robotics and Automation Letters*, vol. 8, no. 4, pp. 2397–2404, Apr. 2023, ISSN: 2377-3774. DOI: 10.1109/lra.2023.3246839. [Online]. Available: <http://dx.doi.org/10.1109/LRA.2023.3246839>.